

Cloud Strategy & Azure Adoption - Case Study Solution

Task 1: Cloud Adoption Reasoning

Organizations should move to cloud to reduce infrastructure costs, improve scalability, enable faster deployments, and enhance monitoring and analytics capabilities.

Cloud solves problems like limited scalability, high maintenance cost, slow deployment cycles, and lack of centralized monitoring.

However, cloud does not automatically solve poor application design or inefficient processes.

Task 2: Azure Selection Analysis

Azure is suitable due to strong integration with Microsoft tools, enterprise-grade security, high scalability, strong developer ecosystem, hybrid cloud support, and compliance offerings.

Task 3: Role-Based Cloud Usage Mapping

Role	Usage	Business Benefit
Developers	Build & deploy apps	Faster delivery
IT Admins	Manage infrastructure	Operational efficiency
Security Team	Monitor threats	Improved security
Executives	View dashboards	Better decisions
Data Team	Analyze data	Insights generation

Task 4: Architecture Decision

Architecture includes web front-end, authentication service, application layer, database, monitoring system, and analytics engine.

Categories: Compute, Storage, Networking, Security, Monitoring, Analytics.

Data Flow: User → Frontend → Auth → App → Database → Analytics → Dashboard.

Task 5: Decision Justification

Azure provides business value through cost optimization, technical flexibility, scalable infrastructure, and pay-as-you-go pricing.

Task 6: Risk Awareness

1. Data security risks – Mitigation: encryption and IAM.
2. Vendor lock-in – Mitigation: multi-cloud strategy.
3. Downtime risks – Mitigation: redundancy and backups.